

APPLICATION OF SYSTEM DYNAMICS MODELS TO THE CHICKEN MARKET IN BOLIVIA

I. INTRODUCTION

System dynamics models were introduced in the 1960s and were developed by J. Forrester at MIT. Since then, they have been applied in various branches of science and technology, and particularly in economics. As their name implicitly suggests, these models make it possible to represent complex systems by transforming their elements into an arrangement of flows and stocks, using control variables and their interrelationships to reflect different interactions through relatively simple equations. These equations lead to the construction of systems of differential equations, considering that a stock is the integral resulting from the difference between inflows and outflows, which in turn represent the growth rate of that integral. In a physical analogy, this rate of change is represented by a derivative. The present document is not intended to constitute a treatise on system dynamics; rather, it emphasizes the use of these tools to construct a model of the systemic functioning of the chicken meat market. Through calibration and further improvement, this model may become a tool that can be used through simulation as a laboratory for policy experimentation.

II. DESCRIPTION OF THE SYSTEM

Chicken meat consumption in Bolivia is estimated to exceed 45 kg per person per year. This figure is obtained by calculating the number of broiler chickens produced, between 240 and 260 million per year according to SENASAG data, and considering an average carcass weight, that is, slaughtered animal weight, of 2.1 kg. This level of consumption is considered one of the most significant in the region. According to data from the Latin American Poultry Association, the main consumers of poultry meat are Brazil, with 47 kg per person per year; Venezuela, with 41 kg per person per year; Argentina, with 40.5 kg; and Peru, with 39 kg.

Chicken meat represents the highest consumption of animal protein among the Bolivian population, followed by beef, with annual per capita consumption of 20 to 22 kg. In overall terms, annual chicken meat consumption amounts to approximately 500,000 tons, with an estimated market value of around USD 850 million, not including the commercial egg market. In terms of per capita food consumption among the Bolivian population, chicken ranks third, after potatoes, at approximately 100 kg per person per year, and milk, at 60 liters per person per year.

The main factors that have influenced the substantial increase in poultry meat consumption among the Bolivian population in recent years are the following:

- Increase in per capita income.
- Relative price of chicken meat per kilogram compared with its immediate competitor, beef, considering that this has been influenced by state policies and

measures related to supply, as well as by efforts to keep the prices of the main inputs, maize, soybean, and sorghum, at reasonable levels for producers.

- Improvement in productivity levels, resulting from better poultry production management, which has achieved feed-to-meat conversion levels comparable to international standards, as well as an average rearing period of 45 to 50 days.
- High nutritional value of the product, among other factors.

According to data from the AEMP and poultry farmers' associations, chicken meat consumption has the following distribution:

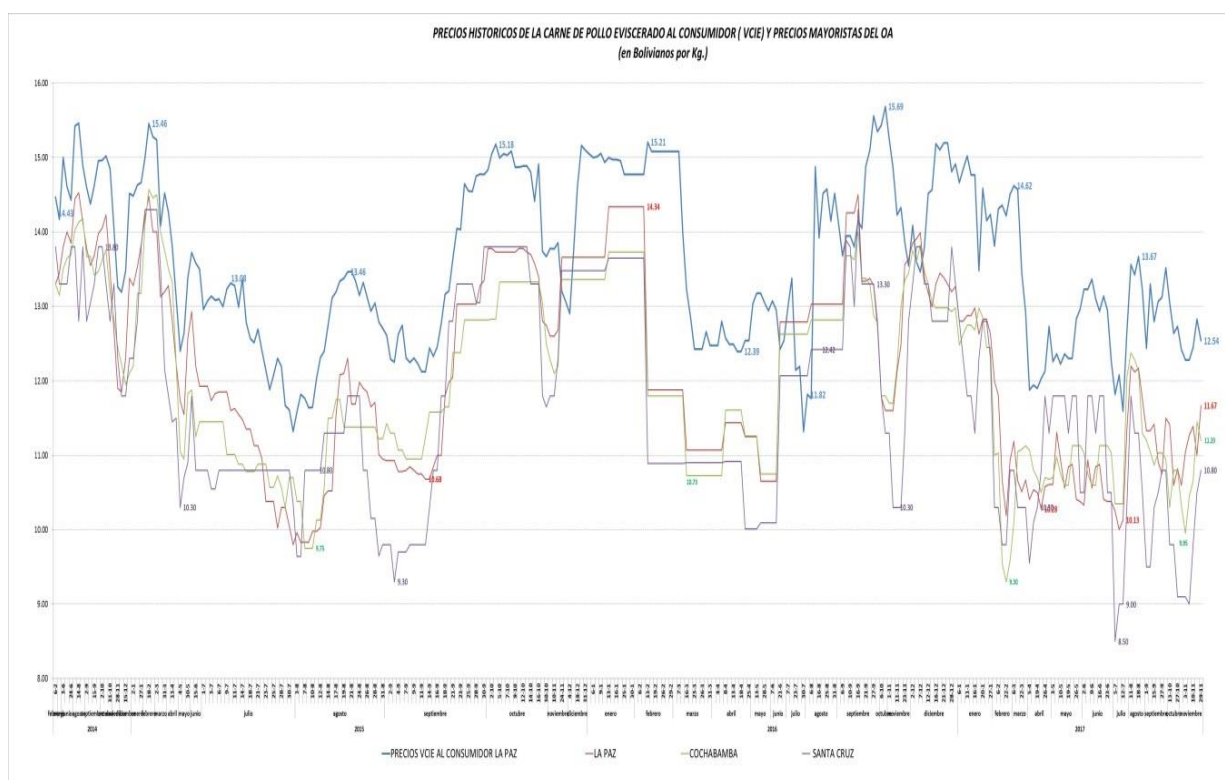
La Paz	45%
Cochabamba	19%
Santa Cruz	24%
Tarija	2%
Oruro	2%
Sucre-Potosí	3%
Beni-Pando	1%

Source: Prepared by the author based on data from the AEMP and poultry farmers' associations.

With regard to La Paz, the main market is undoubtedly the city of El Alto.

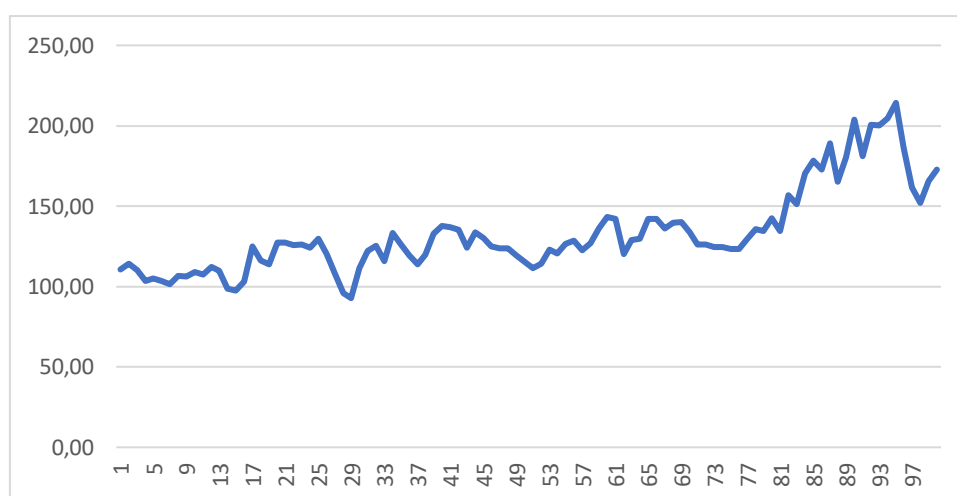
Poultry meat prices, which are approximately 20% of the price of beef, have historically shown substantial variation, implying high price volatility, as shown in Figures 1 and 2.

Figure 1. Variation in chicken prices at intervals of less than one week during the 2014-2018 periods (volatility that has not changed at present)



Source: Prepared by the author based on information from the OAP and the DGDC, VCIE-MDPyEP.

Figure 2. Monthly variation in the chicken price index at intervals of less than one week during the 2018-2026 periods

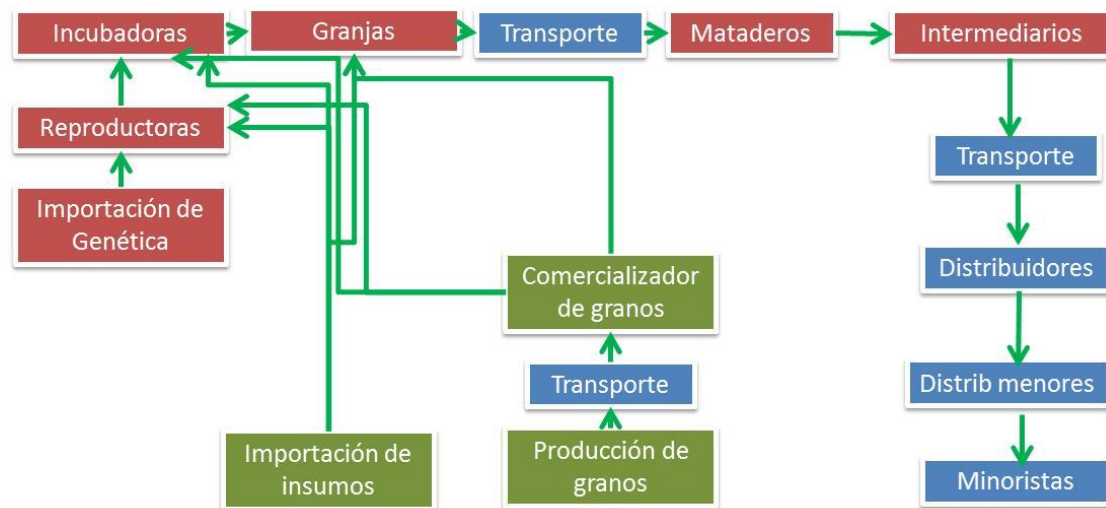


Source: Prepared by the author based on INE data.

As can be observed, volatility persists and an upward trend is evident, the latter due to the inflation experienced by the country in 2025.

With respect to commercialization, the following figure schematically presents the chicken meat production chain, considering the links from the production and importation of inputs, primary production, slaughter, transportation, and wholesale and retail commercialization.

Figure 3. Actors involved in the production and commercialization of poultry meat and its required inputs



Source: Prepared by the author.

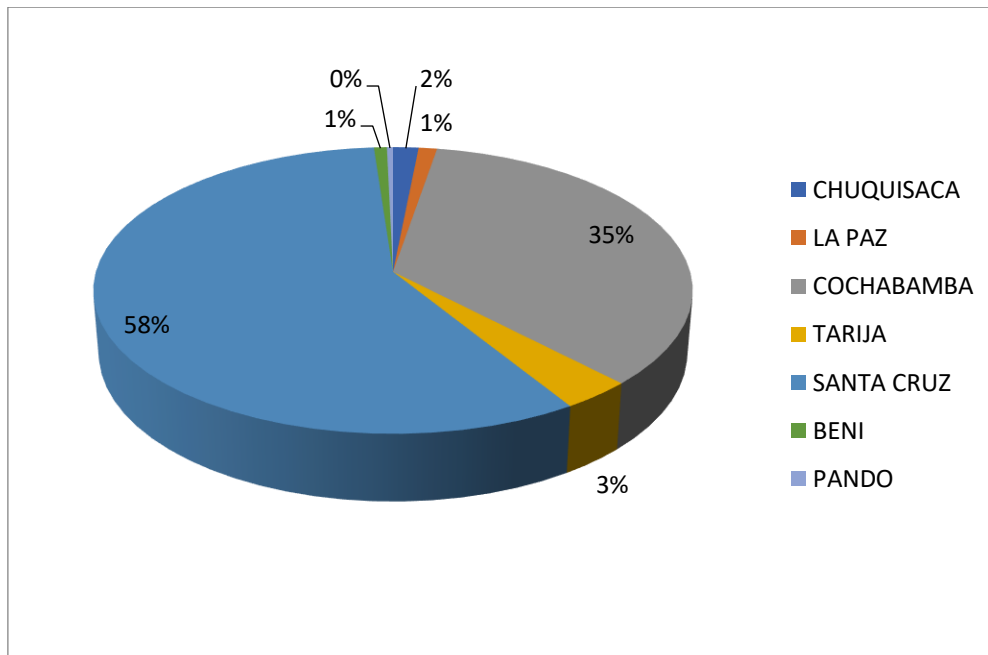
It should be emphasized that, on the supply side, fewer than 10% of producing companies hold a market share exceeding 60%. These companies have also successfully integrated other links of the production chain, such as feed production, the importation of genetic material, and the provision of fertile eggs and day-old chicks. The remaining supply space is covered by many small producers, most of whom depend on the acquisition of day-old chicks from large producers, as well as on inputs supplied by major producers in the industry.

The genetic material, managed under patent-based systems, is imported from Brazil and Argentina through the acquisition of fertile eggs from grandparent stock. These birds, in turn, produce layer birds, which, under appropriate technical management, generate the fertile eggs destined for incubators and the production of day-old chicks, which constitute the basis of primary poultry production on different farms.

This first stage, from the importation of grandparent stock to the production of fertile eggs intended for hatcheries for the production of day-old chicks, involves highly complex veterinary and technical management. For this reason, there are not many companies dedicated to this activity; currently, approximately 21 companies operate in this area.

Finally, with regard to national production, the following figure shows the distribution of production by region.

Figure 5. Chicken meat production in the different departments of Bolivia



Source: Prepared by the author based on INE data.

III. OBJECTIVE

To develop a model of the chicken meat market in Bolivia using system dynamics in order to identify the structural aspects that determine the volatility and high variation of product prices.

No studies using system dynamics have been identified to date on this topic.

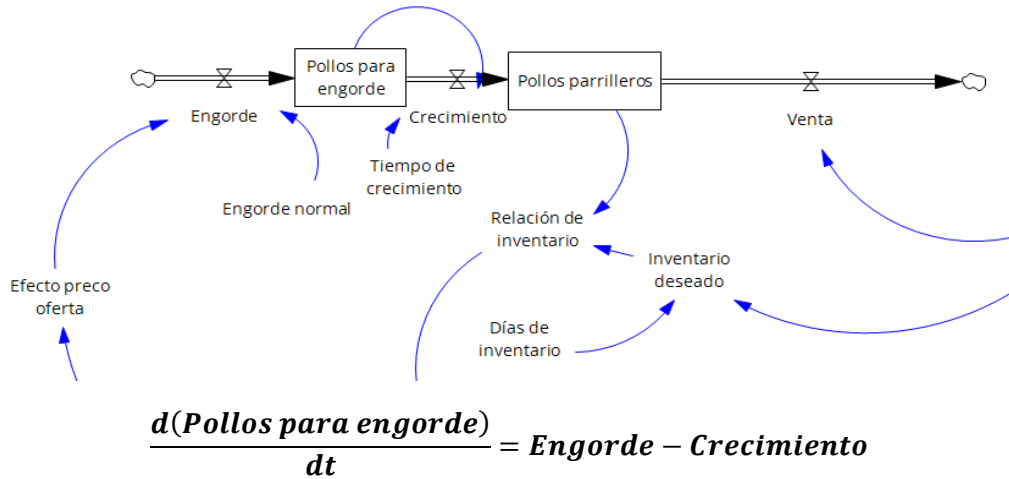
IV. MODEL CONSTRUCTION

The system dynamics model considers blocks and sub-blocks in order to proceed in a structured manner. It uses arrangements of flows, represented by arrows that depart from and arrive at rectangles; stocks or levels, represented by rectangles, which define the state of the system; variables and constants expressed as text; and arrows that connect constants, variables, and flows corresponding to the equations of the model. Within this framework, the model is divided into the following components:

Chicken production

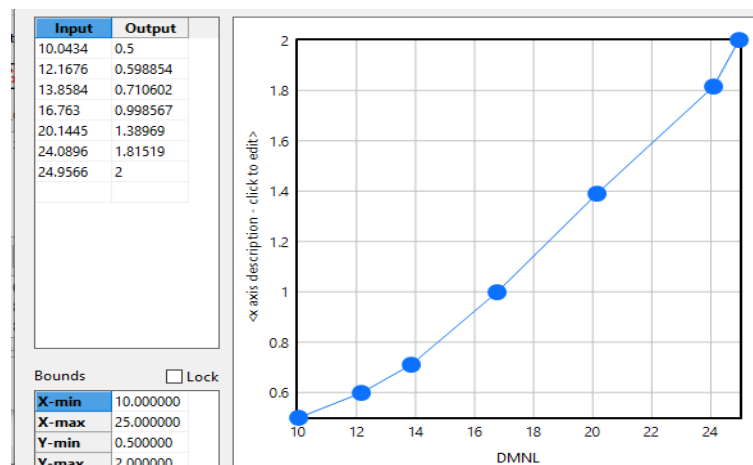
This block of the model simulates the supply of broiler chickens, including the placement of chicks, a decision made by actors based on market price expectations, and their conversion over 45 days from placement to market availability. The model includes two stocks, although more intermediate stocks could be included depending on the required degree of adjustment, that is, material delays of different orders, as shown in the figure.

Figure 6. Modeling of chicken population production



Fattening is equal to the product of normal fattening, considered as a placement of 1.5 million birds per day, and a fattening factor that depends on the market price. This factor is used as a proxy for the price elasticity of supply and was modeled with an adjustable graphical function, shown below. The initial stock values considered were 90 million for the fattening stock and 2 million for broiler chickens, calibrated for an equilibrium situation in order to test market distortions.

Figure 7. Price relationship with the supply-effect factor in the placement of chickens for fattening



This indicates that higher expectations regarding the chicken price would create a greater incentive to place chickens, which, as will be discussed later, is balanced on the demand side.

The equation governing the behavior of the broiler chicken stock is the following:

$$\frac{d(\text{Pollos parrilleros})}{dt} = \text{Crecimiento} - \text{Venta}$$

Growth is equal to the state of the variable representing the population of chickens for fattening divided by 45, which is the average time in days for the bird's growth cycle from the placement of day-old chicks to their delivery to slaughter centers.

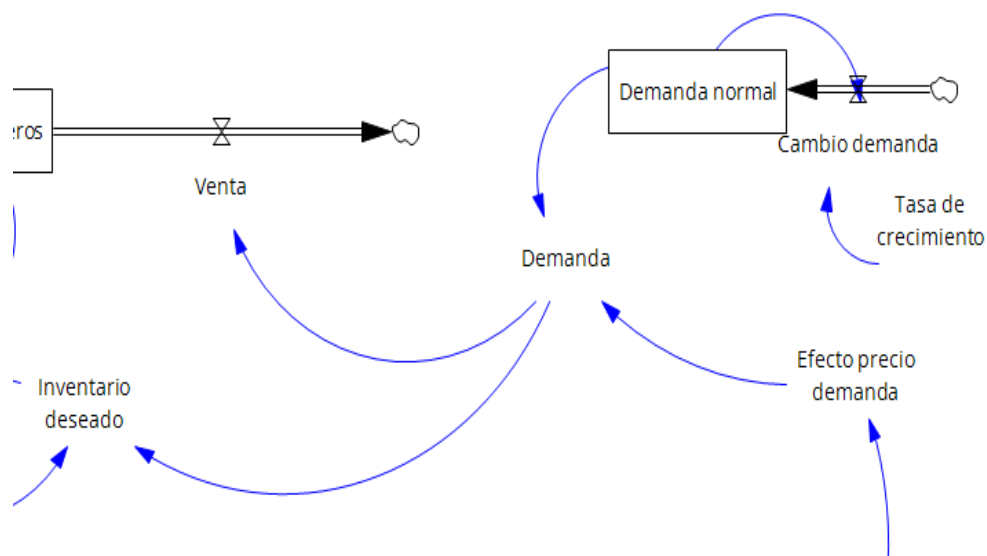
Sales are associated with demand-side variables, which will be discussed below.

Chicken demand

For this block, demand is modeled as a normal demand stock that considers growth associated with the vegetative growth of the population according to the latest census, 1.4% annually, adjusted on a daily basis because that is the time unit used in the model.

The following figure shows the situation described.

Figure 7. Modeling of chicken meat demand



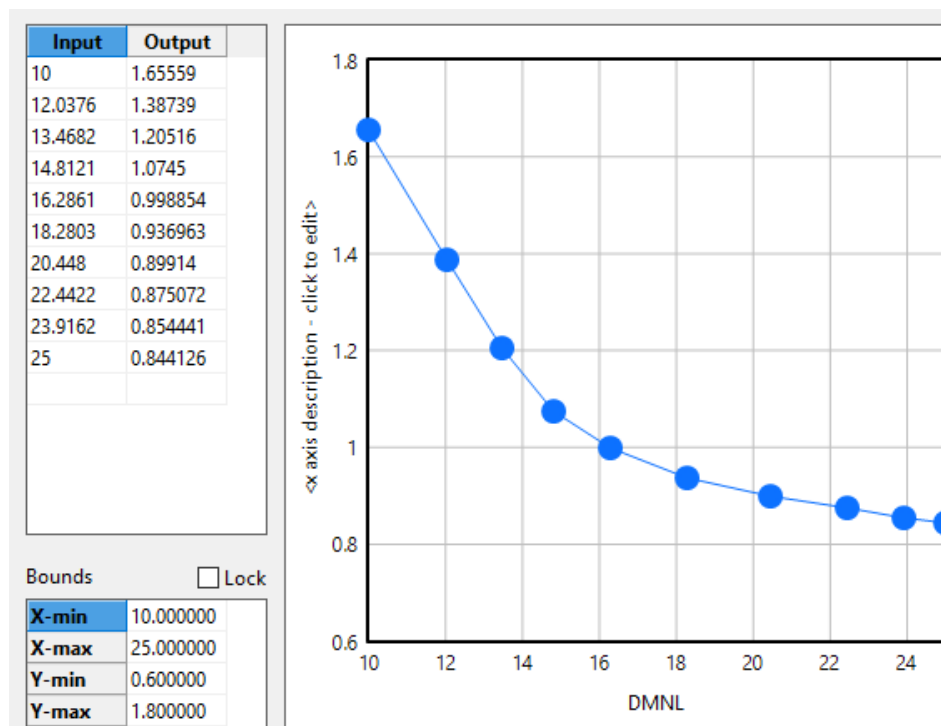
Normal demand is modeled as follows:

$$\frac{d(\text{Demanda normal})}{dt} = \text{cambio de la demanda}$$

The change in demand is equal to the product of the growth rate and the level of normal demand. The rate is equal to 1.4%/365, considering an initial normal demand of 1.5 million chickens per day.

Real demand is equal to the value of the normal demand level multiplied by a price-related demand factor that uses an adjustment curve associated with the price elasticity of demand, which in practice is relatively inelastic, as shown below.

Figure 8. Price relationship with the price-effect factor on demand



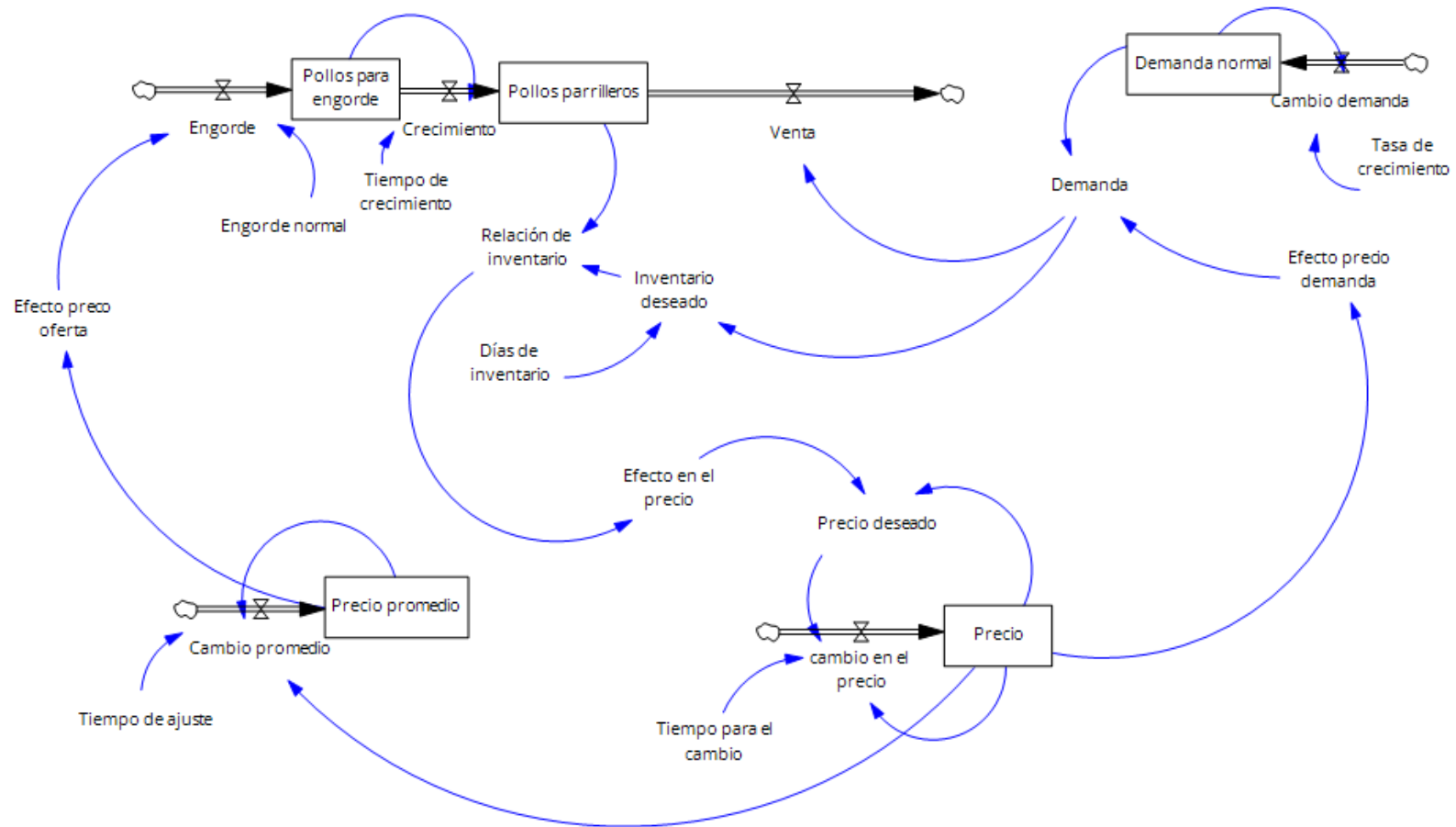
As can be observed, it has a characteristic and slope opposite to those corresponding to the effect of price on supply, meaning that the higher the price, the lower the demand.

Price formation

The following figure presents the integrated model, emphasizing that the stock corresponding to price constitutes the integrating element between supply and demand. It is also related to the evolution of the broiler chicken inventory, since in reality prices react to inventory levels as a direct transmission mechanism of the interaction between supply and demand. In addition, prior to the interaction of price with actors' decisions to modify supply, namely the placement of chickens, an information delay is introduced, considering an average decision-making period of 5 days for price attenuation.

The following figure shows the situation described above.

Figure 9. Complete model with the price stock as the articulating element between supply and demand



Price as a stock has the following equation:

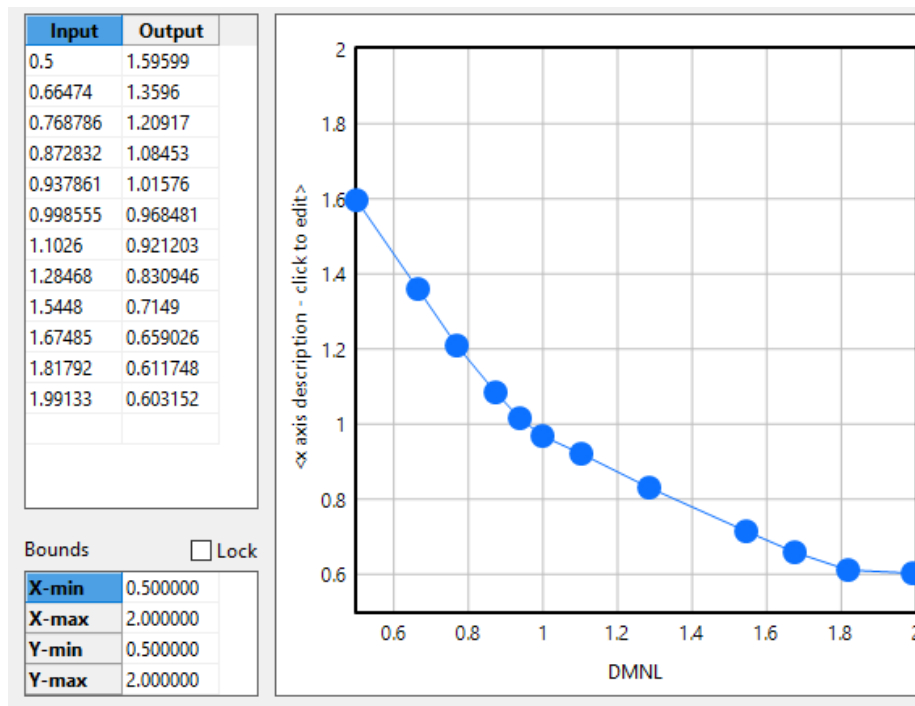
$$\frac{d(\text{Precio})}{dt} = \text{cambio del precio}$$

The change in price is equal to the desired price, determined by the interaction of inventory, minus the current price, divided by the adjustment time.

The desired price corresponds to the inventory ratio. This ratio is established as the quotient between the broiler chicken stock and the desired inventory, which is in turn equal to current demand multiplied by the coverage time, considering a factor of 1.1 because the product is sold on a daily basis.

The nonlinear relationship between the ratio of the broiler chicken inventory and the desired inventory is captured as shown in the following figure, taking into account that a lower value, that is, a decline in actual inventory, produces an effect of increasing the price.

Figure 9. Influence of inventory evolution on price



Finally, the price that interacts with supply and governs the behavior of chicken farmers has an information-delay effect, modeled as follows:

$$\frac{d(\text{Precio promedio})}{dt} = \text{Precio actual} - \text{precio promedio}$$

V. DISCUSSION

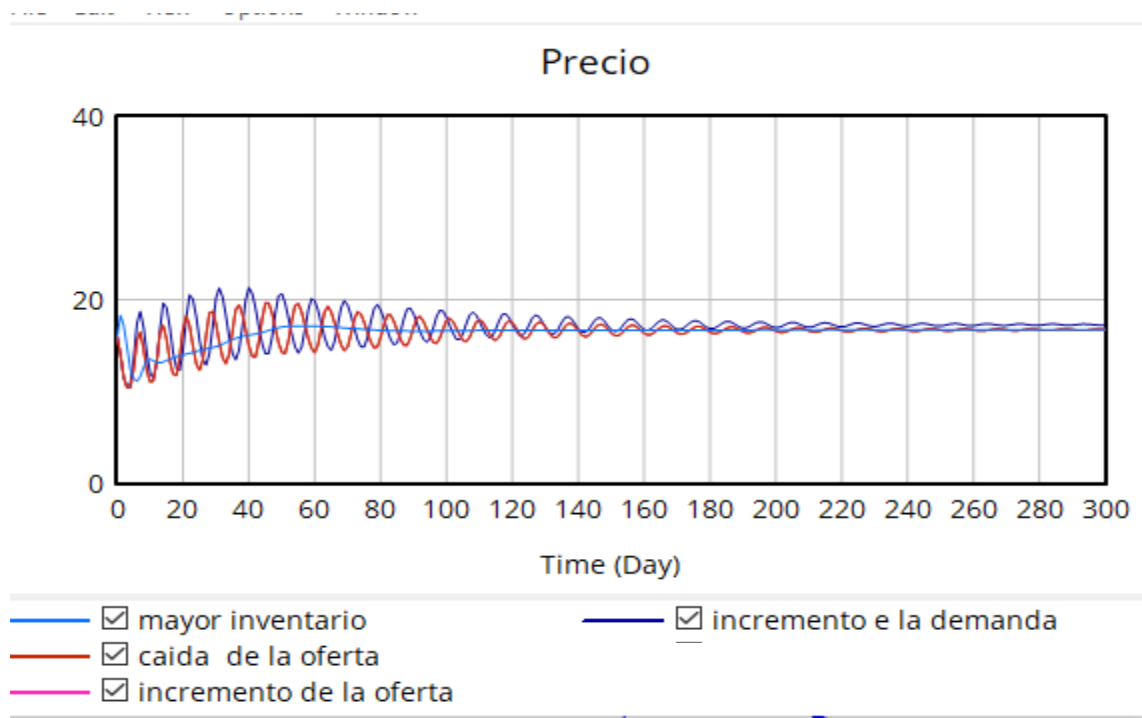
According to the model data, and including distortion factors such as an increase in demand or a decline in supply involving a sudden drop in production over a short period, the results show price oscillations in both cases. Such situations commonly occur in the Bolivian economy, either because of long holidays, such as the Urkupiña festival in Cochabamba in August, road blockades or other conjunctural situations that affect production, or increases in demand mainly related to seasonal factors or, more recently, to significant price increases in a competing product. These oscillations occur in the market, as indicated in previous figures in this document. The decline in production and the increase in supply are modeled using a pulse function with a specific duration.

Since these distortions occur at different points in time, in reality the constant oscillation of chicken prices is what is usually perceived in the market situation.

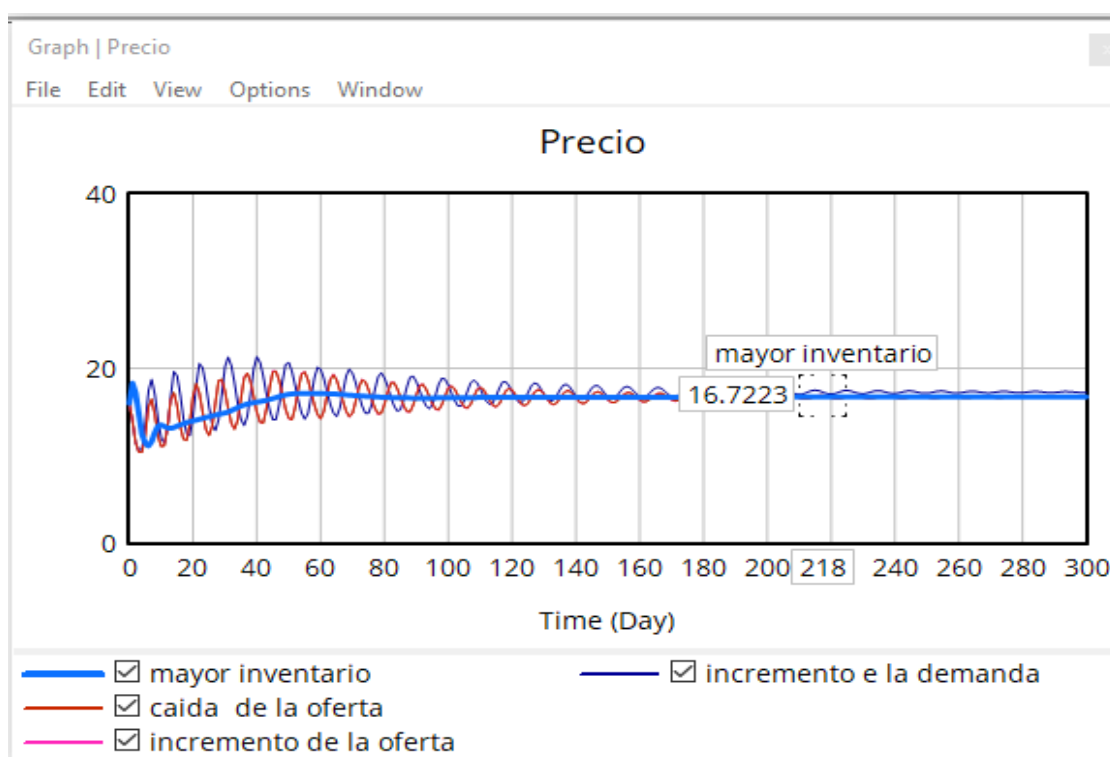
From the standpoint of system structure, the oscillations are mainly based on material delays, such as the time elapsed between the placement of chickens and their entry into the market. This has been modeled with two inventories in order to keep the model simple, although additional inventories could be included to generate third- and fourth-order delay functions. The oscillations are also based on information delays, particularly those related to price formation and to the stock used to model the average price on the basis of which productive actors decide to place chickens, according to their expectations on the supply side.

The results of the simulations are shown in the following figures.

Figures 10 and 11. Influence of inventory evolution on price



Likewise, the following figure highlights a more stable result associated with a larger product inventory in the market. However, the country does not have adequate storage conditions for a longer stock coverage period, since this would require cold-chain systems appropriate for a perishable product.



VI. CONCLUSIONS AND RECOMMENDATIONS

A model using system dynamics tools has been applied to the chicken market in Bolivia. The model reproduces price oscillations in the market, mainly due to the interaction of material and information delays, as well as physical inventories that practically provide daily coverage, considering that the product is highly perishable and requires a cold chain and freezing systems for longer storage periods.

Finally, it should be noted that this model can be improved and that it fulfills the objective of proposing a simple flow relationship that models the observable characteristics of the market. However, future versions would require adjustments based on a larger amount of data in order to achieve greater precision and test different policies.